

PREDICTIVE MENTAL HEALTH ANALYSIS

A Data-Driven Study on Academic Stress and Psychological Defense
Mechanisms among College Students.

EXECUTIVE SUMMARY & STUDY COHORT

OBJECTIVE

This study evaluates the mental well-being of 95 student respondents using the Perceived Stress Scale (PSS-10). Our goal was to uncover hidden patterns of stress that standard surveys often miss.

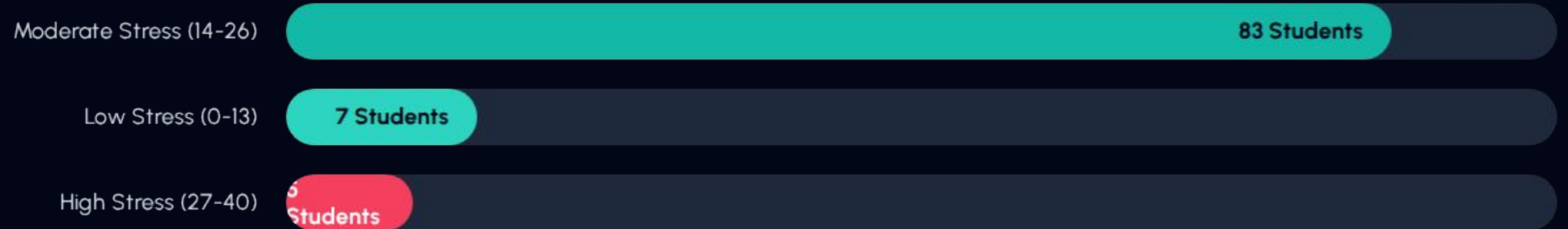
- Total Respondents: 95 Students
 - Metric: PSS-10 Scale (0-40)
 - Focus: Stress Levels & Anomaly Detection
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STRESS DISTRIBUTION

Quantifying the immediate psychological state of the respondent group.

STRESS LEVEL DISTRIBUTION (N=95)



87% of students are currently in the "Moderate" zone, suggesting a high baseline level of academic pressure.

THE MACHINE LEARNING FRAMEWORK



RANDOM FOREST

Utilized 100 decision trees to identify "valid" vs "anomaly" response paths with high predictive accuracy.



FEATURE SELECTION

Analyzed 10 psychological parameters to determine which questions were most predictive of emotional state.



ANOMALY DETECTION

Audited behavior for "Ego Defense" mechanisms—where verbal coping claims contradict crisis data.

| THE ANOMALY DETECTION AUDIT

83.2%

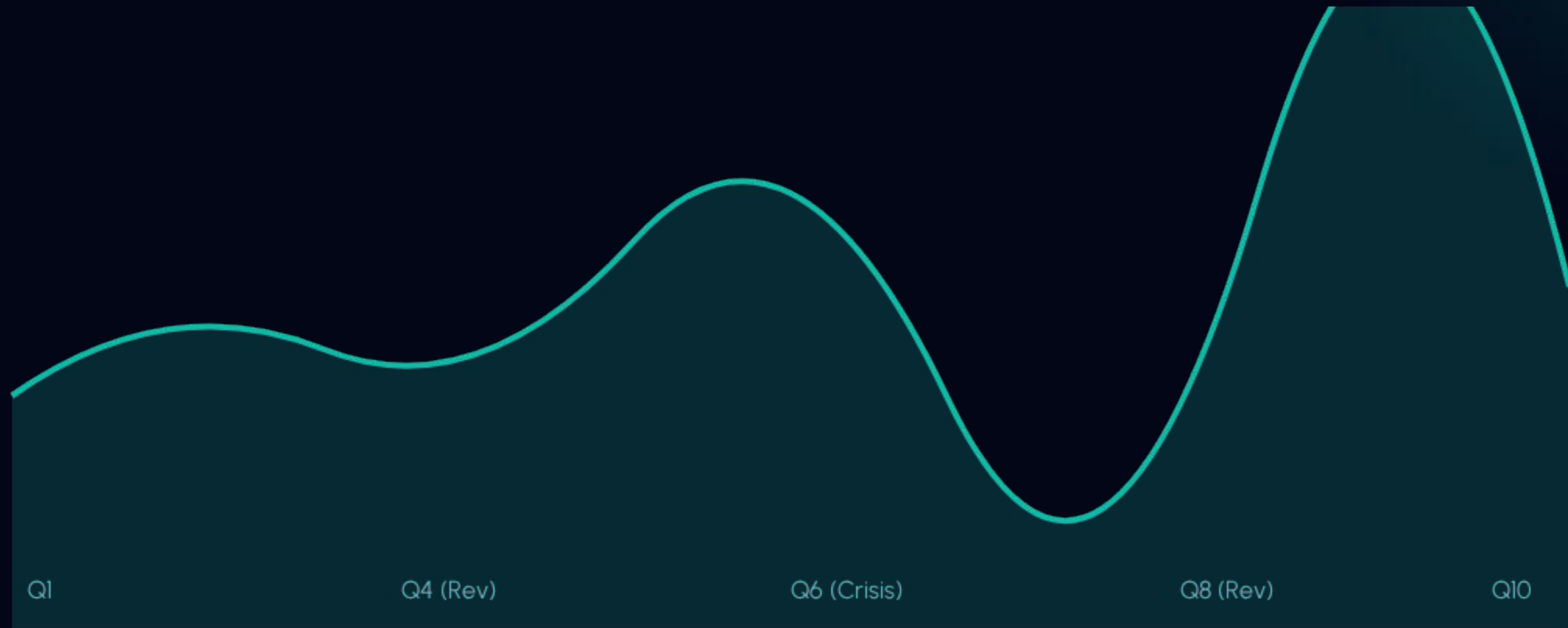
DENIAL RATE

SIGNIFICANT BEHAVIORAL BIAS

Out of 95 respondents, 79 individuals displayed extreme response jumps or contradictions, suggesting they are under-reporting their actual distress levels.

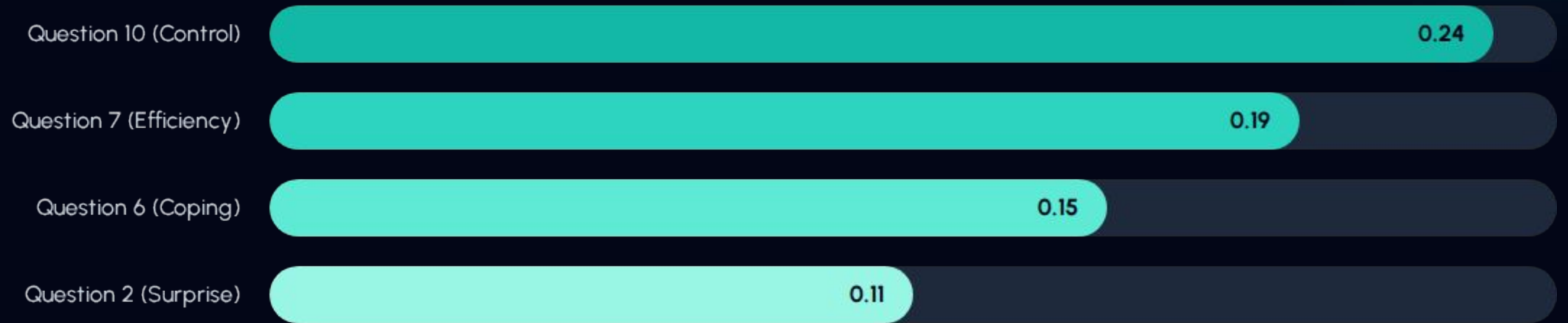
This "Denial" mechanism is a common psychological response to academic stigma.

EMOTIONAL RESPONSE FLOW ANALYSIS



Visualization of the collective "Dynamic Path" of 95 respondents across the PSS-10 scale.

PREDICTIVE AI: FEATURE IMPORTANCE



Questions regarding "Feeling in Control" (Q10) were the strongest indicators of emotional stability according to the Random Forest model.

THE EGO DEFENSE PARADOX

CONTRADICTIONARY RESULTS

Students reported being "Very Capable" of coping (Q5), yet simultaneously "Overwhelmed" by crisis (Q6).

$$\text{Score}_{\text{Defense}} = 4 - |Q_6 - Q_5|$$

Average Ego Defense Score: **2.92 / 4.00**. This indicates a high level of psychological masking.

AUDIT: THEORETICAL VS REAL PATH

Category	Theoretical Prob.	Actual Survey Data	Interpretation
Path Variation	9,765,625 Possibilities	93 Unique Routes	High Individuality
Normal Flow	Low Deviation	16 Students	Consistent/Stable
Extremes (Denial)	Jumps > 2 Levels	79 Students	Psychological Masking

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"Stress is real, but the silence and denial driven by social stigma are often more dangerous than the pressure itself."

— **Mental Health Data Insights**

Final Conclusion: 83.2% of our students need a safe space to admit vulnerability.

DISCLAIMER

I am not a licensed psychologist or mental health professional. This analysis was conducted strictly as a personal project using data science methodologies, building upon my experience during my Master's (S2) thesis where I deeply explored psychological theories. All conclusions, indicators, and models presented here are entirely my own personal theories and hypotheses, which could be subjective or incorrect. This should not be used as a professional psychological diagnosis.

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